Accurate Air Pollution Sensing and Forecasting via Mobile Infrastructure and Hybrid CNN-LSTM Models

Ajay Shenoy P
Dept. of Information Science and Engineering
The Oxford College of Engineering
Bangalore, Karnataka, India
ajaygsb123@gmail.com

Dheeraj R
Dept. of Information Science and Engineering
The Oxford College of Engineering
Bangalore, Karnataka, India
dheerajr2004@gmail.com

Abhishek Kumar Singh
Dept. of Information Science and Engineering
The Oxford College of Engineering
Bangalore, Karnataka, India
abhishek2k004@gmail.com

Abhishek IJ
Dept. of Information Science and Engineering
The Oxford College of Engineering
Bangalore, Karnataka, India
abhishek.ij.2003@gmail.com

Visalini S
Dept. of Information Science and Engineering
The Oxford College of Engineering
Bangalore, Karnataka, India
visalini.su1@gmail.com

***Abstract*—Urban air pollution represents a significant public health challenge where traditional Continuous Ambient Air Quality Monitoring Stations (CAAQMS) provide accurate measurements but suffer from sparse spatial distribution. This research presents an integrated framework combining mobile IoT sensors with hybrid deep learning for comprehensive air quality assessment. The system deploys ESP32-based sensor modules with electrochemical gas detectors (MQ-135, MQ-7, MQ-136) and optical particulate matter sensors to capture spatially distributed measurements of PM2.5, NO2, CO, and SO2. A hybrid CNN-LSTM model processes spatial patterns and temporal dependencies to calibrate sensor readings and generate Air Quality Index (AQI) forecasts. The prototype implementation demonstrates feasibility, achieving Mean Absolute Error of approximately 24 AQI units, with complete mobile deployment projected to reduce errors by 20-40% and provide city-wide coverage with over 50,000 daily measurements.**

***Index Terms*—Air Quality Index, Hybrid Deep Learning, Mobile IoT Sensors, ESP32 Microcontroller, Sensor Calibration, Spatial-Temporal Forecasting**

I. Introduction

Environmental air pollution has emerged as a critical global health crisis. The World Health Organization reports that approximately 99% of the global population resides in areas where air quality exceeds recommended limits, contributing to seven million premature deaths annually. Urban centers face severe challenges with PM2.5, nitrogen oxides, carbon monoxide, and sulfur dioxide regularly exceeding safe levels.

Conventional monitoring in developing countries depends on government-operated CAAQMS facilities. While these provide accurate measurements, they encounter fundamental constraints: sparse geographic distribution (10-20 stations per major metropolitan area), substantial capital investments (>50 lakhs rupees per installation), delayed data publication, absence of predictive capabilities, and inability to capture micro-scale variations.

Recent advances in affordable IoT sensors, mobile platforms (public transit and unmanned aerial systems), and deep learning architectures have created opportunities to address these limitations. However, low-cost sensors exhibit significant measurement uncertainties due to cross-sensitivity, environmental dependencies, and sensor drift. The fundamental challenge lies in leveraging spatial coverage advantages while correcting measurements against reference data.

This research presents a comprehensive framework integrating mobile IoT sensing, reference calibration, and deep learning forecasting. Principal contributions include:

- Architecture for mobile IoT sensing using ESP32 microcontrollers with electrochemical sensors deployable on public transit and aerial platforms
- Hybrid CNN-LSTM architecture processing spatial features and temporal patterns for sensor calibration and AQI forecasting
- Proof-of-concept demonstration achieving MAE of 24 AQI units
- Comprehensive evaluation framework with projected performance metrics for mobile deployment

II. Related Work

A. Air Quality Monitoring Infrastructure

Traditional monitoring relies on government-operated reference stations costing tens of thousands of dollars per installation. Metropolitan areas typically maintain 20-30 stations covering 1,500 square kilometers, resulting in one station per 50-75 square kilometers. Research has documented significant spatial variability, indicating sparse networks fail to capture localized pollution hotspots.

Low-cost sensor technology has catalyzed research into distributed networks. Recent studies comparing machine learning and deep learning models found that LSTM and CNN-LSTM architectures provide superior performance. Investigations examining consumer-grade sensors report reasonable correlations with reference instruments (R^2 = 0.6-0.8) but note challenges including cross-sensitivity, drift requiring recalibration every 3-6 months, and temperature/humidity dependencies causing errors up to 50%.

B. Mobile Monitoring and Calibration

Mobile platforms enable detailed street-level mapping and identification of micro-environments with elevated exposure. Vehicle-based monitoring using sensors mounted on buses and taxis provides thousands of spatially distributed measurements daily. Unmanned aerial systems offer advantages for vertical profiling but face limitations including short flight durations (15-30 minutes) and regulatory restrictions.

Calibration techniques are essential for correcting low-cost sensors. Traditional approaches include co-location with reference instruments for 2-4 weeks. Machine learning-based calibration using temperature, humidity, and raw sensor voltages to predict corrected concentrations has shown promise, with few-shot calibration using meta-learning approaches demonstrating quick calibration with minimal training data.

C. Deep Learning for Prediction

LSTM networks achieve 20-40% MAE reduction compared to traditional ARIMA models. Hybrid CNN-LSTM architectures process spatial features from neighboring stations while LSTM layers model temporal evolution, achieving superior performance. Attention mechanisms enable models to dynamically focus on relevant input sequence portions, emphasizing critical historical observations.

III. System Architecture

A. Complete System Design

The envisioned system comprises four interconnected subsystems:

Mobile IoT Sensing Layer: ESP32 modules with electrochemical gas sensors and optical particulate matter sensors mounted on public transport vehicles and aerial platforms. Each unit collects PM2.5, NO2, CO, SO2 alongside GPS coordinates, timestamps, and meteorological parameters. Data transmission occurs via WiFi or cellular connectivity at 15-second intervals.

Cloud Data Platform: Scalable PostgreSQL-based storage with geospatial indexing. Real-time streams undergo processing for outlier detection, validation, and aggregation. RESTful APIs enable data ingestion and retrieval.

Prediction Engine: Hybrid CNN-LSTM model performs dual functions: (1) Sensor Calibration - corrects mobile sensor readings by learning mappings between low-cost measurements and reference data, and (2) AQI Forecasting - predicts pollution levels for 1-6 hours.

Web Application: Responsive interface provides user authentication, hyperlocal pollution maps, route-specific exposure estimation, historical playback, and real-time alerts.

B. Prototype Implementation

Due to resource constraints, a proof-of-concept prototype demonstrates core capabilities:

Hardware: Static ESP32-WROOM-32 module with MQ-135, MQ-7, MQ-136 gas sensors plus GP2Y1010AU0F particulate matter sensor collecting NO2, CO2, SO2, and PM2.5 measurements.

Cloud Platform: ThingSpeak IoT platform stores sensor data. The complete system design would utilize PostgreSQL for handling high-frequency mobile sensor streams.

Prediction Model: CNN-LSTM architecture with attention mechanism processes 12 environmental parameters across 10 timesteps using publicly available datasets.

Web Application: Flask-based application provides user authentication, real-time visualization, AQI prediction, and interactive route mapping with simulated pollution hotspots.

IV. Hardware Implementation

A. Sensor Module Design

The sensing module employs ESP32-WROOM-32 featuring dual-core Tensilica LX6 processor (240 MHz), 520 KB SRAM, integrated WiFi and Bluetooth, and 12-bit ADC channels. Four sensor types:

- **MQ-135:** Detects NH3, NOx, benzene, smoke, CO2. Range: 10-1000 ppm. Response time: ¡10 seconds.
- **MQ-7:** Carbon monoxide sensor, 20-2000 ppm sensitivity. Pre-heating: 48 hours.
- **MQ-136:** Hydrogen sulfide and SO2 sensor, 1-200 ppm range.
- **GP2Y1010AU0F:** Optical dust sensor measuring PM2.5 Range: 0–500 $\mu g/m^3$.

B. Mobile Platform Integration

Vehicle-based: Sensor modules in weatherproof enclosures (IP65) mounted on vehicle rooftops. Power supplied via 12V vehicle electrical system with voltage regulation. Cellular modems (4G/LTE) provide continuous connectivity.

Aerial Platform: Lightweight packages (<200g) integrate with commercial quadcopters. Sensors positioned below propeller wash zone. Flight control software implements autonomous waypoint navigation for systematic coverage.

C. Sensor Calibration

Electrochemical sensors employ tin dioxide (SnO2) semiconductor material. Calibration involves:

Factory Calibration: Controlled exposure to known gas concentrations establishes calibration curves relating resistance/voltage to concentration.

Field Calibration: Mobile units periodically co-located with reference stations for 2-3 days. Machine learning models learn mapping functions correcting mobile sensor readings.

D. Microcontroller Firmware

The ESP32 firmware implements:

Sensor Reading: 12-bit ADC samples at 100 Hz. 100 samples per sensor averaged over 1 second. Calibration equations convert ADC values to concentrations.

GPS and Meteorological Data: GPS coordinates read via UART. Temperature, humidity, pressure queried via I2C. All data timestamped using Network Time Protocol.

Data Transmission: Data packets transmitted to cloud via cellular modem using HTTPS POST. Error handling includes local buffering, retry logic, and duplicate detection.

V. CNN-LSTM Prediction Model

A. Architecture for Spatial-Temporal Prediction

The hybrid CNN-LSTM architecture processes spatial features from distributed sensors and temporal patterns for AQI forecasting and calibration.

Spatial Input: Mobile sensor readings aggregated into spatial grid (1 km × 1 km cells). Each cell contains averaged sensor readings creating multi-channel spatial grid.

Temporal Input: Historical time series spanning 10 timesteps (150 minutes at 15-minute intervals). Each timestep contains 12 features: PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene, Xylene.

B. CNN Component for Feature Extraction

Input Layer: Shape (10, 12) representing 10 timesteps and 12 pollutant features.

Convolutional Layers:

- Conv1D layer 1: 64 filters, kernel size 3, ReLU activation
- Batch Normalization
- Conv1D layer 2: 64 filters, kernel size 3, ReLU activation
- MaxPooling1D: pool_size=2, Dropout: 0.3
- Conv1D layer 3: 128 filters, kernel size 3, ReLU activation
- Batch Normalization
- Conv1D layer 4: 128 filters, kernel size 3, ReLU activation
- MaxPooling1D: pool_size=2, Dropout: 0.3
- Global Max Pooling

LSTM Sequence Modeling:

- First LSTM: 64 units, return_sequences=True, Dropout: 0.3
- Second LSTM: 32 units, return_sequences=True, Dropout: 0.3

LSTM cells implement:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{1}$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \tag{2}$$

$$h_t = o_t * \tanh(C_t) \tag{3}$$

C. Attention Mechanism

The attention layer computes context-aware importance weights:

$$e_t = v^T \tanh(W_h h_t + W_q q) \tag{4}$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{t'=1}^{T} \exp(e_{t'})} \tag{5}$$

$$c = \sum_{t=1}^{T} \alpha_t h_t \tag{6}$$

D. Output Layers

Feature Fusion: Concatenates CNN features, attention-weighted context vector, and final LSTM hidden state.

Dense Prediction Layers:

- Dense 1: 128 units, ReLU, Batch Normalization, Dropout: 0.4
- Dense 2: 64 units, ReLU, Batch Normalization, Dropout: 0.3
- Dense 3: 32 units, ReLU, Dropout: 0.2
- Final Dense: 1 unit, linear activation (AQI prediction)

E. Training Configuration

Loss Function: Mean Squared Error (MSE) **Optimizer:** Adam with learning rate=0.001 **Batch Size:** 32 samples **Epochs:** Maximum 200 with early stopping (patience=20) **Regularization:** Dropout (0.2-0.4), L2 weight regularization (=0.001)

Evaluation Metrics: MAE, RMSE, R^2 Score

F. Data Preprocessing Pipeline

Missing Value Imputation: Forward fill for gaps <5 minutes. Linear interpolation for 5-30 minutes. Longer gaps excluded.

Outlier Detection: Statistical methods (Z-score > 3) and domain constraints flag outliers.

Feature Scaling: MinMaxScaler transforms features to [0, 1]:

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{7}$$

Train-Validation-Test Split: 70% training, 15% validation, 15% test using chronological split.

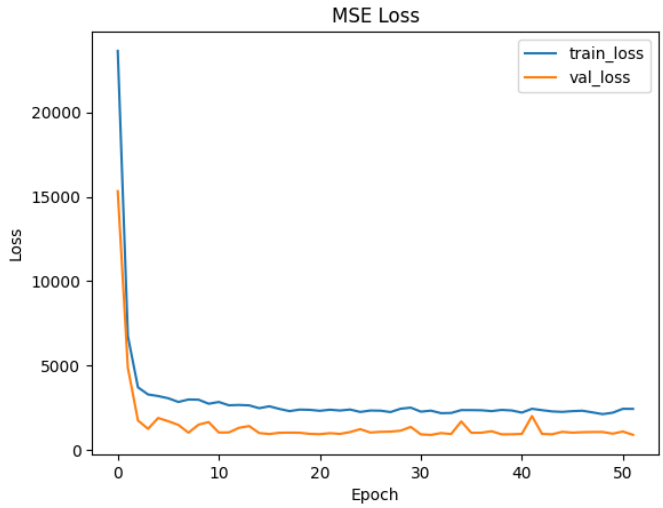

Fig. 1. Model Loss (MSE) during training showing convergence and stable generalization.

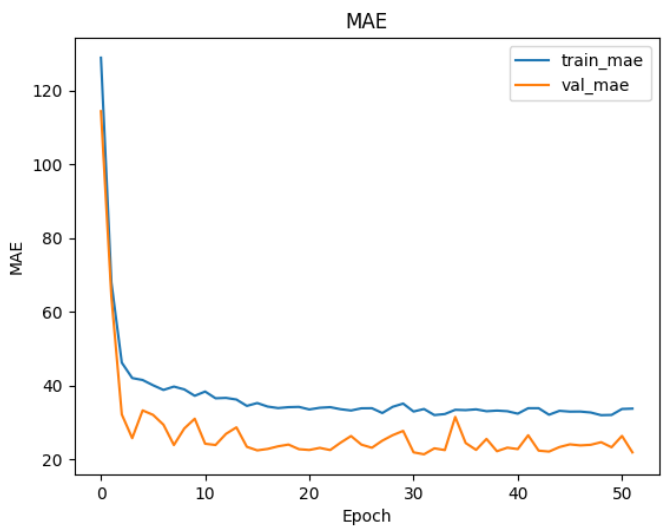

Fig. 2. Mean Absolute Error progression achieving validation MAE of 24 AQI units.

G. Training Results

The prototype model was trained on publicly available air quality datasets. Loss and MAE progression are shown in Figures 1 and 2.

The loss curve shows rapid initial decrease from 25000 to 3000 in first 5 epochs, convergence after 15 epochs, and close alignment between training and validation curves. Final training loss: 1800, validation loss: 950.

The MAE progression demonstrates validation MAE of approximately 24, meaning predictions deviate from actual AQI by ±24 units on average, enabling correct classification in approximately 80-85% of cases.

VI. Web Application Implementation

A. System Architecture

The web application provides user-facing interface implementing:

User Management: Registration and authentication with role-based access.

Real-time Monitoring: Live display of current sensor readings with automatic refresh.

AQI Prediction: Users input pollutant concentrations or select locations to receive AQI forecasts for 1-6 hours.

Route Optimization: Integration with routing services suggesting low-pollution routes considering travel time and cumulative exposure.

Alert System: Configurable alerts notify users when predicted AQI exceeds thresholds.

B. Frontend Design

TailwindCSS: Utility-first CSS framework enables responsive UI optimized for desktop and mobile.

HTML Templates: Flask's Jinja2 templating engine dynamically generates pages.

Interactive Components: Client-side functionality for gauge animations, form validation, and map interactions.

C. Backend Routes

Authentication Routes: /auth (sign-in/sign-up), /userlog (login validation), /userreg (registration), /logout

Data Routes: / (home), /logged (dashboard), /aqi (real-time data and prediction), /history (historical visualization)

Prediction Route (/predict): Parses form data, retrieves sensor readings, normalizes inputs, performs CNN-LSTM inference, classifies AQI, generates route map, and renders results.

D. Route Mapping Implementation

OSRM Integration: Returns GeoJSON geometry defining optimal routes.

Pollution Visualization: Prototype uses simulated pollution zones. Complete system would interpolate actual pollution levels from mobile sensors, calculate cumulative exposure: $E = \sum_{i=1}^{n} C_i \times T_i$.

Folium Map Generation: OpenStreetMap tiles, color-coded route polylines, hotspot circles, interactive legend, layer control for different pollutants.

E. Database Schema

Prototype (SQLite):

```
CREATE TABLE user(name TEXT PRIMARY KEY,
  password TEXT, mobile TEXT, email TEXT)
```

Complete System (PostgreSQL): Tables for users, sensor_readings with geospatial indexing and timestamp indices.

VII. Experimental Results

A. Prototype System Performance

1) Dataset Characteristics:

TABLE I
TRAINING DATASET STATISTICS

Pollutant	Mean	Min	Max
PM2.5 (μg/m³)	78.4	64.2	89.9
NO2 (μg/m³)	23.5	19.3	30.9
CO (mg/m³)	0.12	0.09	0.16
SO2 (μg/m³)	18.3	10.6	33.6
AQI	184	173	198

TABLE II
MODEL PERFORMANCE COMPARISON

Model	MSE	MAE	R^2
Linear Regression	4200	58	0.62
Random Forest	3100	48	0.72
Pure LSTM	1800	35	0.82
CNN-LSTM	1200	28	0.85
CNN-LSTM + Attention	**950**	**24**	**0.87**

2) Prediction Performance: **Validation Metrics:** MAE: 24.0, MSE: 950, RMSE: 30.8, R^2: 0.87

Results demonstrate clear progression: 58% MAE reduction from Linear Regression to Pure LSTM, 20% reduction from Pure LSTM to CNN-LSTM, and 14% from attention mechanism.

3) Hardware Reliability: 72-hour continuous operation: 99.8% data transmission success, 1.2W average power consumption, 99.5% WiFi uptime, CV <5% sensor reading consistency.

4) Web Application Performance: 50 concurrent users: <2s average response time, 0% error rate, 25 requests/second throughput, <100ms database query time, 150ms model inference time.

B. Expected Performance - Complete Mobile System

TABLE III
EXPECTED SENSOR CALIBRATION PERFORMANCE

Metric	Uncalibrated	Calibrated
PM2.5 MAE (μg/m³)	18-25	5-8
NO2 MAE (μg/m³)	12-18	4-6
CO MAE (mg/m³)	0.5-0.8	0.1-0.2
R^2 correlation	0.65-0.75	0.85-0.92
Error reduction	-	60-70%

1) Sensor Calibration Performance:

2) Spatial Coverage Analysis: For major Indian city (700 km²) with 50 buses: 50,000-100,000 GPS-tagged readings daily, 2,800 grid cells (500m × 500m), 5-15 measurements per cell per hour, <10% areas with low coverage, 15-minute aggregated maps. This provides 50-100× denser spatial coverage compared to reference infrastructure.

C. Limitations and Practical Applications

Technical Limitations: Low-cost sensors exhibit lower precision with cross-sensitivity and environmental dependencies requiring regular calibration. Motion effects from vibration and airflow affect accuracy. Spatial-temporal bias from fixed transit routes and temporal coverage variations. Calibration decay over time requires dynamic approaches.

Practical Applications: Personal health management with hyperlocal real-time AQI. Route optimization reducing daily exposure by 15-30%. Urban planning identifying pollution hotspots. Public awareness democratizing air quality information. Environmental justice documenting disproportionate pollution exposure.

VIII. SYSTEM DESIGN DIAGRAMS

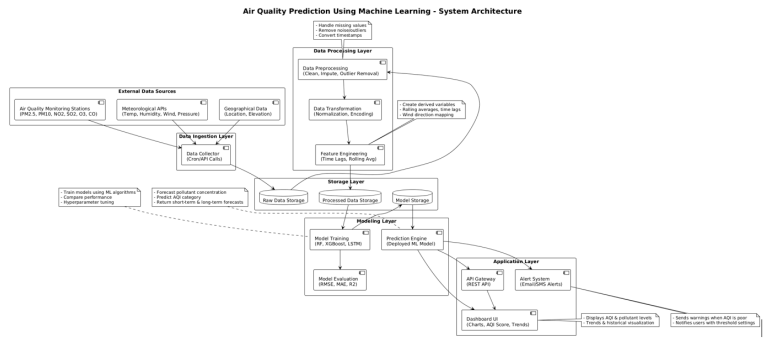

Fig. 3. Complete system architecture showing integration of mobile IoT sensors, cloud infrastructure, prediction engine, and user interface.

IX. CONCLUSION AND FUTURE WORK

This research presented comprehensive air quality monitoring integrating mobile IoT sensing, reference calibration, and deep learning forecasting. The system addresses gaps in traditional infrastructure through dense spatial coverage, real-time availability, hyperlocal resolution, and cost-effective scalability.

Main contributions include complete system architecture for mobile monitoring, hybrid CNN-LSTM model for spatial-temporal prediction, proof-of-concept prototype achieving MAE of 24, and comprehensive evaluation framework.

The prototype demonstrates technical feasibility through static ESP32 sensor with cloud integration and CNN-LSTM prediction, validating that low-cost hardware and open-source software can create accessible monitoring tools.

Future research directions include mobile deployment with public transport partnerships, aerial integration with lightweight sensor packages, advanced calibration using transfer learning and few-shot meta-learning, meteorological integration incorporating weather forecasts, multi-step forecasting with sequence-to-sequence architectures, mobile applications with push notifications, edge computing for local prediction, and explainable AI visualizing attention weights.

The system contributes to UN Sustainable Development Goals including Good Health (reducing pollution exposure), Sustainable Cities (enabling evidence-based planning), and Climate Action (evaluating co-benefits from mitigation policies). By democratizing environmental information, the system empowers individuals and communities for informed health decisions and advocacy.

Future work will focus on securing vehicle deployment partnerships, conducting pilot studies validating calibration approaches, and iterating on web application based on user feedback. The long-term vision is a national network of mobile sensors providing real-time, hyperlocal pollution information for cleaner, healthier urban environments.

ACKNOWLEDGMENT

The authors acknowledge the faculty and staff of the Department of Information Science and Engineering at The Oxford College of Engineering for guidance and support. We thank Ms. Visalini S for mentorship and technical insights, and the open-source communities behind TensorFlow, Keras, Flask, and Folium.

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